SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2
Assessment Tool Weightage: 50%

QUESTIONS: 2 **Total Marks:50**

Annexure A

Technical Presentation

Criteria	***Technical Content Accuracy(3)***	***Problem Identification(2.5)***	***Use of Tools(2)***	***Communication(1.5)***	***Professionalism(1)***	***Total(10)***
Weightage	*30%*	25%	20%	20%	10%	*100%*
Q1						
Q2						

Code of Conduct:

I _M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M.Pranay*

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Topic 1: Decision Support Systems (DSS)

🎯 Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations

Topic 2: Operational Systems vs DSS

🎯 Focus Areas:

1. OLTP vs OLAP concepts
2. Key differences (data, users, purpose)
3. Performance comparison
4. Use-case scenarios
5. Integration between systems

Topic 3: Architecture of Data Warehouse

🎯 Focus Areas:

1. Data warehouse definition
2. Three-tier architecture
3. ETL process
4. Data marts & metadata
5. Benefits in decision-making

Topic 4: OLAP and Data Cubes

🎯 Focus Areas:

1. OLAP concept
2. Data cube structure
3. OLAP operations (slice, dice, roll-up, drill-down)
4. Applications in analytics

Topic 5: Dimensional Data Modeling

🎯 Focus Areas:

1. Fact and dimension tables
2. Star schema vs snowflake schema
3. Schema design process
4. Real-world example

Topic 6: Query Processing & BI Tools

🎯 Focus Areas:

1. Query processing stages
2. Query optimization
3. BI tools overview
4. Reporting and visualization
5. Examples of open-source BI tools

Topic 7: OLAP Application using KNIME

🎯 Focus Areas:

1. Introduction to KNIME
2. Workflow components (nodes, connections)
3. Performing OLAP operations
4. Sample application/demo

5. Report generation

PART B: PRESENTATION RULES & GUIDELINES

◇ 1. Group Formation

1. Each group: 3–5 students
2. Each member must participate
3. Assign roles:
    1. Presenter
    2. Designer (PPT)
    3. Researcher
    4. Demo/Case handler

◇ 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
    1. Definition and importance
2. Core Concepts
    1. Explanation with diagrams
3. Process/Architecture
    1. Step-by-step explanation
4. Real-World Example
    1. Case study or application
5. Tool/Demo (if applicable)
    1. KNIME or BI tool
6. Conclusion
    1. Summary + future scope

◇ 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
    1. Diagrams
    2. Flowcharts

    3. Tables (for comparison)
3. Avoid:
    1. Too much text
    2. Copy-paste content

◇ 4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
    1. Eye contact
    2. Clear voice
    3. Logical flow

ANSWERS

1.Decision Support Systems (DSS)

1. Introduction

A Decision Support System (DSS) is a computer-based information system that helps managers analyze data and make effective decisions. It is used for solving complex and semi-structured business problems.

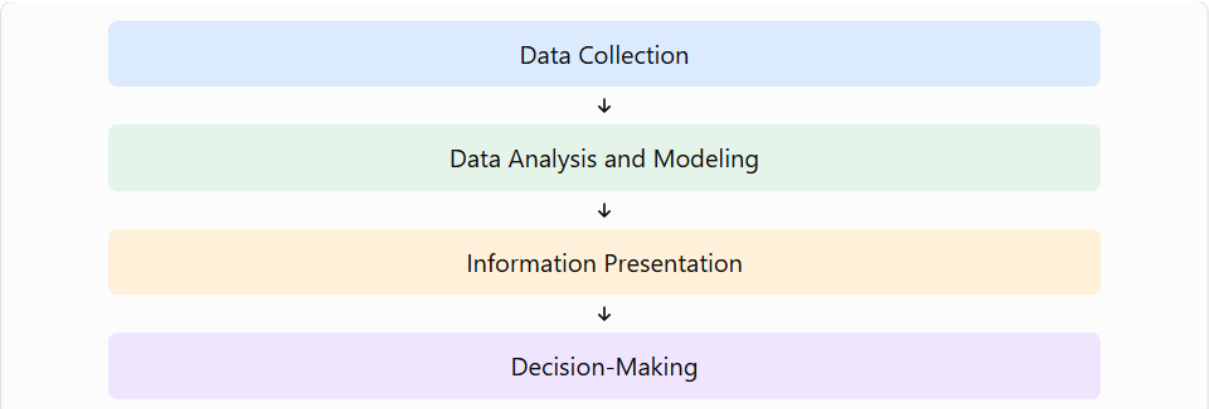
2. Components of DSS

- Data management: Collects and stores data from different sources.
- Model management: Uses mathematical models, forecasting and statistical analysis.
- User interface (UI): Displays information through reports, charts and dashboards.

3. Types of DSS

- Data-driven DSS
- Model-driven DSS
- Knowledge-driven DSS
- Document-driven DSS
- Communication-driven DSS

4. Working process



5. Real-world applications

- Banking: Evaluates loan applications and financial risks.
- Healthcare: Helps hospitals plan resources and analyze patient information.
- Retail: Helps managers forecast demand and maintain stock.

6. Advantages and limitations

Advantages: Improves decision quality, saves time, identifies trends and supports planning.

Limitations: Depends on data quality, can be costly to implement and requires skilled users.

7. Conclusion

DSS transforms raw data into useful information and helps organizations make faster, more informed decisions.

2. Operational Systems vs DSS

1. Introduction

Operational systems manage everyday transactions, while Decision Support Systems analyze data to support business decisions. OLTP and OLAP are the main concepts used to understand this difference.

2. OLTP vs OLAP

- OLTP (Online Transaction Processing): Handles daily transactions, such as payments, orders and deposits.
- OLAP (Online Analytical Processing): Analyzes large volumes of data to identify trends and support planning.

3. Key differences

Feature	OLTP	OLAP / DSS
Purpose	Daily operations	Analysis and decisions
Data	Current, detailed	Historical, summarized
Users	Operational staff	Managers and analysts
Queries	Short and frequent	Complex and analytical
Main workload	Insert, update, delete	Read, aggregate, compare
Example	ATM transactions	Bank profit analysis

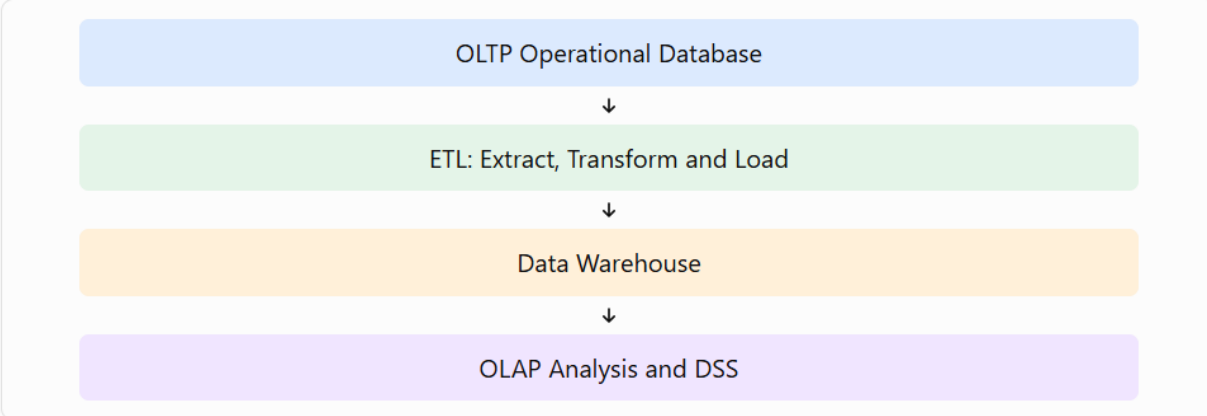
4. Performance comparison

OLTP is optimized for fast transaction processing and maintaining data consistency. OLAP is optimized for complex queries and aggregating large datasets. Their performance depends on database design, indexing, hardware and workload.

5. Use-case scenarios

- Banking: OLTP records deposits and withdrawals; OLAP analyzes customer transactions and loan trends.
- Retail: OLTP records laptop purchases; OLAP compares sales by brand, region and month.

6. Integration between systems



7. Conclusion

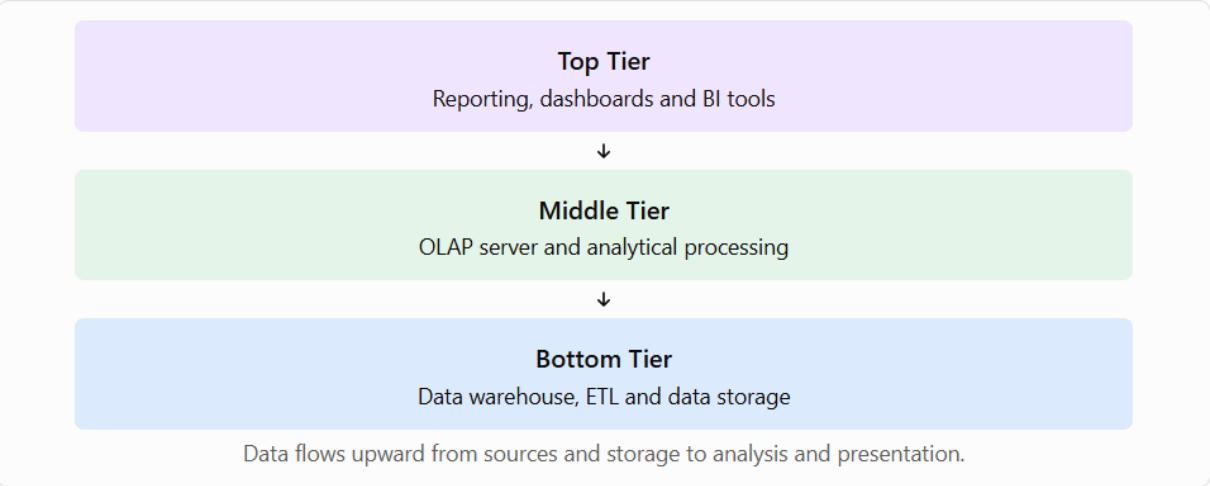
OLTP supports daily business activities, whereas OLAP helps analyze business performance. Together, they support both operational efficiency and strategic decision-making.

3. Architecture of a Data Warehouse

1. Introduction

A data warehouse is a centralized repository that stores integrated data from multiple sources for reporting, analysis and decision-making.

2. Three-tier architecture



3. ETL process

- Extract: Collects data from databases, files and business applications.
- Transform: Cleans, validates and converts data into a consistent format.
- Load: Stores the transformed data in the data warehouse.

4. Data marts and metadata

- Data marts: Smaller, subject-specific data repositories for departments such as sales, finance and marketing.
- Metadata: Describes the structure, meaning, source and format of stored data.

5. Benefits in decision-making

- Integrates information from multiple sources.
- Stores historical data for trend analysis.
- Improves reporting and consistency.
- Supports strategic planning and forecasting.

6. Real-world example

An electronics retailer collects laptop sales data from multiple stores, cleans it through ETL and stores it in a data warehouse. Managers use dashboards to compare sales by brand, price and location.

7. Conclusion

A data warehouse provides reliable, organized and historical information for effective business intelligence and decision-making.

4. OLAP and Data Cubes

1. Introduction

OLAP stands for Online Analytical Processing. It enables users to analyze data from multiple dimensions, such as time, product and location.

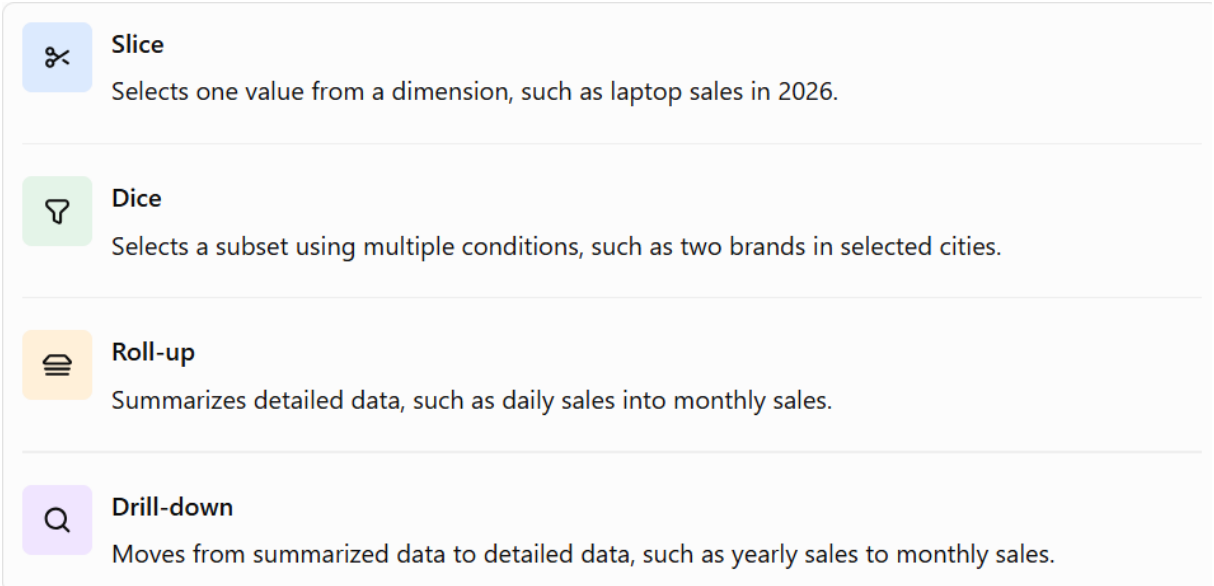
2. Data cube structure

A data cube represents data using dimensions and measures.

- Dimensions: Time, product, location and customer.
- Measures: Sales amount, quantity sold and profit.

For example, laptop sales can be analyzed by brand, month and city.

3. OLAP operations



4. Applications in analytics

- Sales and revenue analysis.
- Financial performance analysis.
- Inventory management.
- Customer purchasing trend analysis.
- Regional and product comparisons.

5. Real-world example

A laptop retailer uses OLAP to identify which laptop brand generates the highest revenue in each city and during each month.

6. Conclusion

OLAP and data cubes simplify multidimensional analysis, helping managers identify patterns, compare performance and make informed decisions.

5. Dimensional Data Modeling

1. Introduction

Dimensional data modeling is a technique for organizing data in a data warehouse so that business users can easily query, analyze and report information.

2. Fact and dimension tables

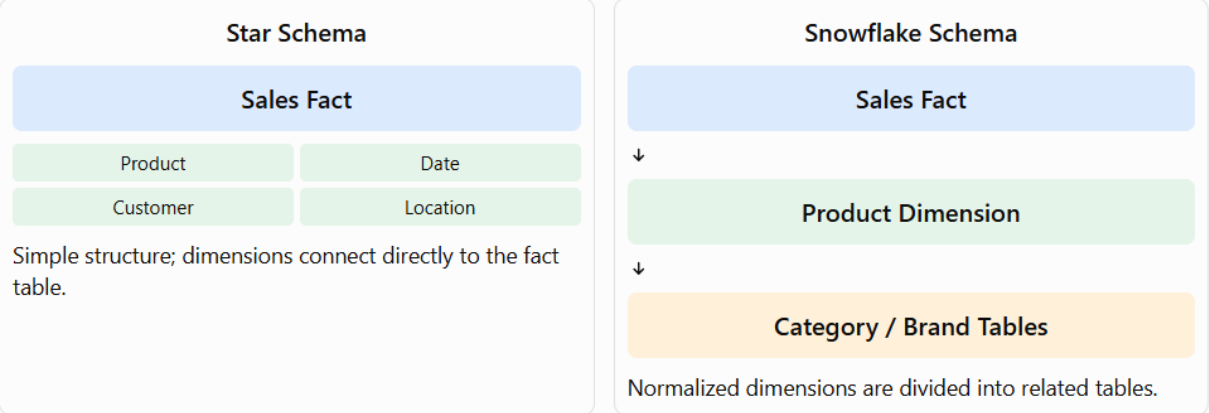
- Fact table: Stores numerical business measurements, such as sales amount, quantity and profit.
- Dimension table: Stores descriptive attributes, such as product name, date, customer and location.

Example:

Fact table: Sales	**Value**
Product ID	P101
Date ID	D01
Quantity	2
Sales amount	₹80,000

The Product ID and Date ID connect the fact table to the corresponding dimension tables.

3. Star schema vs snowflake schema



Feature	Star schema	Snowflake schema
Structure	Simple	More complex
Dimensions	Usually denormalized	Normalized
Joins	Fewer	More
Querying	Usually simpler	May require more joins
Storage	May use more space	May reduce redundant data

4. Schema design process

1. Identify the business process, such as laptop sales.
2. Define the level of detail (grain), such as one row per sales transaction item.
3. Identify numerical facts and measures.
4. Identify dimensions such as product, date, store and customer.
5. Define keys and relationships.
6. Choose a suitable schema and validate it with sample queries.

5. Real-world example

An electronics store creates a sales fact table connected to product, date and store dimensions. It can then calculate total revenue by brand, compare monthly sales and identify high-performing stores.

6. Conclusion

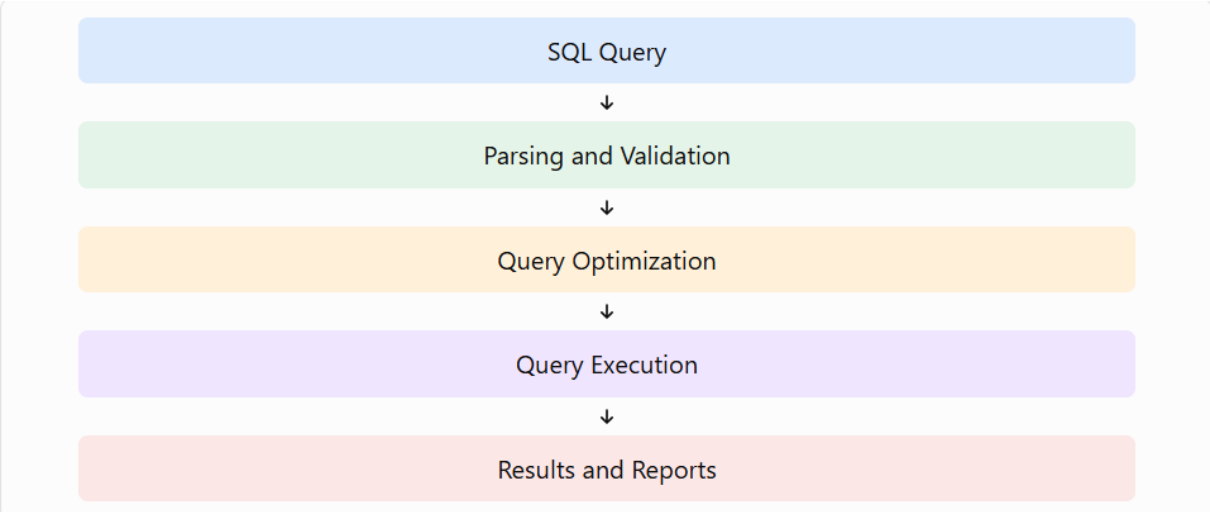
Dimensional modeling makes warehouse data easier to understand, query and analyze. Star and snowflake schemas provide different ways to organize fact and dimension tables.

6. Query Processing and BI Tools

1. Introduction

Query processing is the process of converting a database query into an executable plan and retrieving the required results. Business Intelligence (BI) tools use these results for reporting, visualization and decision-making.

2. Query processing stages



3. Query optimization

Query optimization selects an efficient way to execute a query.

Common techniques include:

- Using indexes to retrieve records faster.
- Filtering unnecessary records early.
- Selecting only required columns.
- Choosing efficient join operations.
- Reviewing query execution plans.

4. BI tools overview

BI tools help users collect, analyze and present business data.

Examples include:

- Power BI: Interactive reports and dashboards.
- Tableau: Data visualization and analytics.
- KNIME: Visual workflows for data preparation and analysis.
- Metabase: Open-source business intelligence and dashboards.

- Apache Superset: Open-source data exploration and visualization.

5. Reporting and visualization

BI tools present information through tables, bar charts, line charts, pie charts and dashboards. These visualizations help identify trends and compare business performance.

6. Real-world example

A laptop retailer runs an SQL query to calculate monthly sales for each brand. A BI tool displays the results in a chart, helping the manager compare brands and plan inventory.

7. Conclusion

Query processing retrieves useful information efficiently, while BI tools convert the results into understandable reports and visualizations for better decisions.

7. OLAP Application Using KNIME

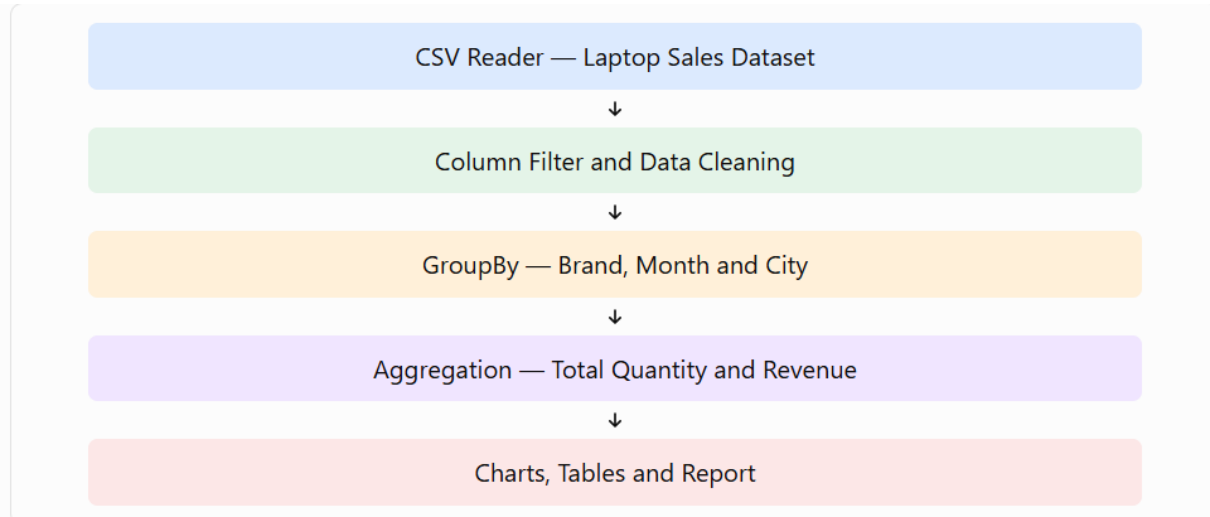
1. Introduction

KNIME is a visual analytics platform that allows users to build data processing and analysis workflows by connecting nodes. It can be used to prepare sales data, perform aggregations and create visual reports.

2. Workflow components

- Nodes: Perform operations such as reading files, filtering data and grouping records.
- Connections: Transfer data between nodes.
- Data tables: Hold the data processed by the workflow.
- Output nodes: Display or export the final results.

3. Sample workflow



4. Performing OLAP operations in KNIME

OLAP operation	Example application
Slice	View laptop sales for 2026 only
Dice	View two brands in selected cities
Roll-up	Summarize monthly sales by year
Drill-down	Examine yearly sales month by month

These operations can be implemented using suitable filtering, grouping and aggregation nodes.

5. Sample application/demo

Consider a dataset containing the following fields:

Brand	Month	City	Quantity	Revenue
Dell	January	Chennai	10	500000
HP	January	Chennai	8	400000
Dell	February	Hyderabad	12	600000
Lenovo	February	Chennai	6	300000

In KNIME:

1. Import the CSV file using the CSV Reader node.
2. Inspect and clean the data.
3. Use GroupBy to group records by brand or month.
4. Calculate total quantity and revenue.
5. Display the results in a table or chart.
6. Export the results or use them in a report.

The figures above are sample data for demonstration, not actual market sales.

6. Report generation

The analysis can produce:

- Brand-wise sales reports.
- Monthly revenue summaries.
- City-wise product comparisons.
- Charts showing sales trends.
- Information to support stock planning.

7. Advantages and limitations

Advantages: Visual workflow design, reusable processes, support for multiple data sources and reduced need for manual coding.

Limitations: Users must understand the workflow and data structure. Some advanced integrations and reporting features may require additional extensions or configuration.

8. Conclusion

KNIME enables users to build visual data analysis workflows and perform OLAP-style operations. It helps transform raw sales data into summaries, charts and useful business information.